THE UNITED REPUBLIC OF TANZANIA
THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT
ANNUAL JOINT EXAMINATION

FORM ONE
031 PHYSICS

Time 2:30 Hours **November 2024**

INSTRUCTION:

- This paper consist of three sections A, B, and C
- Answer ALL questions in all sections
- Whenever necessary use the following constants
  i. Acceleration due to gravity, g = 10 m/s^2 or 10 N/kg
  ii. Density of water = 1 g/cm^3 or 1,000 kg/m^3
  iii. Density of alcohol = 780 kg/m^3
  iv. Density of an unknown cylindrical solid = 1500 kg/m^3

NO.	ATTEMPTED QUESTIONS	MARKS (SCORE)	EXAMINER'S INITIALS
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
TOTAL SCORE			

SECTION A (15 Marks)

1.This question consists of ten (10) multiple-choice questions. Answer ALL questions by writing the letter of the correct answer against the item number.

(i) To measure the length using a metre rule, wrong position of the eye leads to:

A. Parallalx error B. Eye error C. Zero error D. Metre rule

(ii) Which of the following is not a matter?

A. Heat B. Air C. Steam D. Cloud

(iii) A mass of 60 kg weighs 600 N on the Earth and 100 N on the Moon. What is the mass and weight of an object on the Earth if it weighs 50 N on the Moon?

A. 60 kg mass, 600 N weight B. 10 kg mass, 60 N weight

C. 30 kg mass, 300 N weight D. 5 kg mass, 100 N weight

(iv) A piece of metal of volume 0.24 cm^3 and mass 0.72 g has a relative density

A. 3.0 g/cm^3

B. 3.0kg/m^3

C. 0.3

D. 3.0

(v) 480 centimeters is equivalent to ……. metres:

A. 0.48m

B. 4800m

C. 4.8m

D. 0.048m

(vi) The following statements are all true except one. Which one is that?

A. A Physics is capable of answering many questions about nature

B. Physics is used in industry, music and medicine

C. There is no relationship between physics and biology

D. All discoveries in physics have been found and no one is left.

(vii) The ratio of masses of the objects in figure is $m_1/m_2 = 4 / 3$. What is the ratio of densities of the objects d_1/d_2?

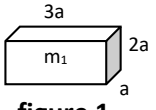


figure 1

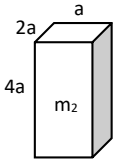


figure 2

A. 9 / 16 B. 8 / 7 C. 1 D. 16 / 9

(viii) The volume of a piece of metal with a mass of 150 g and density of 0.03 g/cm^3 is;

A. $5\times 10^{-3} cm^3$

B. $5\times 10^{1}cm^{3}$

C. $5\times 10^{2}cm^{3}$

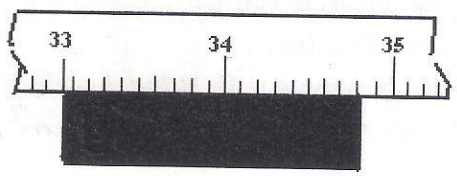
D. $5\times 10^{3}cm^{3}$

(ix) As one goes far away the earth, the density of air:

A. Decreases B. Increases

C. Remain constant D. Becomes greater than its weight

(x) The figure below shows a metre rule being used to measure the length of a wood.



The length of the wood is approximately;

A. 1.8cm B. 2.8 cm C. 34.8cm D. 1.8mm

Answers:

i	ii	iii	iv	v	vi	vii	viii	ix	x

2. Match each item in **LIST A** with a correct response in **LIST B** by writing its letter below the number of the corresponding item in **LIST A** in the table provided.

LIST A	**LIST B**
i) Elasticity ii) Surface tension iii) Osmosis iv) Capillarity v) Evaporation	A) Force on free liquid surface B) Force of attraction between similar molecules C) Ability of a liquid to climb up narrow tubes D) Change of shape of an object to regain its shape E) Movement of molecules through a semi permeable membrane F) Change from liquid to gas at any temperature G) Change from liquid to gas at specific temperature

ANSWERS

List A:	I	ii	iii	iv	v
List B:					

SECTION B (70 Marks)

Answer **ALL** questions from this section

3. (a) What is meant by the term derived unit? Give three examples of derived units.

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_____
_____
_____

(b) Briefly explain the importance of forming hypothesis before doing an experiment.

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(c) Seypalm is a student who is less interested in physics, he claims that physics is USELESS and that is a DEAD SUBJECT. Explain in briefly how SEYPALM can be convinced to learn physics by showing four (4) major reasons for studying physics by any secondary student.

(i)_____
_____
_____

(ii)_____
_____
_____

(iii)_____
_____
_____

(iv)_____
_____
_____

4. (a) Differentiate between first aid and first aid kit**.**

_____
_____
_____

(b) Mention four first aid kit items and its function as showing on the table below. (**8 marks**).

ITEMS	FUNCTION
i.	
ii.	
iii.	
iv.	

5. (a) Differentiate between mass and weight.

(b) Seypalm stands on the scale and it reads 54 kg.

(i) What is the mass of Seypalm?
(ii) What is his weight?

(c) If was to go to another planet where the gravitational force is $2\ N/kg$ what would be his weight and his mass in this planet?

6. (a) State one application of Archimedes' principle in your daily life.

(b) A cylindrical solid of area of cross-section $0.0004\ m^2$ and length $0.30\ m$ is completely immersed in water. Calculate

(i) Weight of solid in S.I. system.
(ii) Upthrust acting on solid in S.I. system.
(iii)Apparent weight of solid in water and alcohol.

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7. (a) A metal pin was observed to float on the surface of pure water. Explain why the pin would sink if a few drops of a soap were added to the water?

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_____
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_____

(b) A spiral spring extended by 10 cm when a force of 0.5 N was applied at its ends. Calculate;

(i) Force constant.
(ii) Force which causes an extension of 20 cm.

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8. (a) Your classmate has an urgent message to relay to his parents. If it were to appear in writing, which methods would you advise him to use. Name four (4)

(i) _____
(ii) _____
(iii)_____
(iv)_____

(b) What should you do in the laboratory in the event of the following situations?

(i) You need to carry out an experiment but there is nobody in the laboratory.

_____

_____

(ii) Your partner is cut by a glass during experiment.

_____

_____

(iii) You want to burn waste paper.

_____

_____

(iv) You have your Kilimanjaro water and you want to drink it.

_____

_____

(v) One of your partners suggests that you carry the apparatus out of the laboratory in order to finish the experiment at home.

_____

_____

9. (a) Describe the way liquids exerts pressure.

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_____

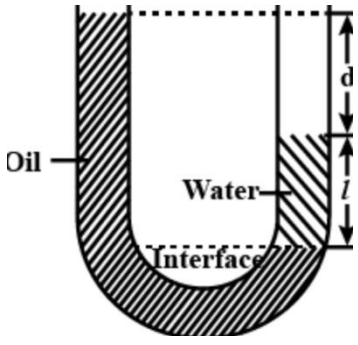
_____

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(b) A U-tube contains two liquids in static equilibrium.
Water of density $\rho_w (= 1000 kgm^{-3})$ is in the right arm, oil of unknown density ρ is in the left arm as shown in the figure



Measurement give $l = 135\ mm$ and $d = 12.5\ mm$. Calculate the density of oil.

SECTION C (15 Marks)

Answer **ALL** questions from this section.

10. (a) Mention instrument used to measure the length of an object to its accuracy value.

(3 marks)

Items	**Accuracy(mm)**
i.	
ii.	
iii.	

(b) Your provided with eureka can, stone, thread, water, small beaker and the mass of stone. Explain on how can you determine the density of stone**. (4 marks**)

(c) show that $1 g/cm^3$ is equal to 1000 kg/m^3**(4 marks)**

d) Draw the Vernier scale and the main scale of 10.64cm. (**4 marks)**